IBRAHIM TIJANI, PhD, P.Eng, PMP

Google Scholar | Scopus | Researchgate | Materials informatics | AI for resilient infrastructure

Canada | +1 (204) 901 2501 | ibrahim.tijani@ubc.ca | ibrahim.tijani04@yahoo.com

PEER-REVIEWED JOURNAL PUBLICATIONS

24 peer-reviewed journal articles | 17 Q1 | 6 Q2 | 23/24 (96%) published in Q1 – Q2 journals

1. **Tijani I.A**, Haroglu H., Wakjira T.G, Abadel A., & Alam M.S (2026). Data-augmented explainable machine learning framework and user-centric tool for classifying failure modes and predicting flexural capacity of FRP-strengthened RC beams. *Structures*. 111776. [Q1].
2. **Tijani I.A**, Kanaan D, Wakjira T.G, Alam M.S, & Haroglu H (2026). Explainable machine learning in concrete technology: A state-of-the-art review with future innovations. *Archives of Computational Methods in Engineering*. 33, 6095-6123. [Q1].
3. **Tijani I.A** & C Jiang. (2025) GUI-based hybrid ML model for predicting ultimate strength of FRP-confined UHPC with CTGAN-augmented data. *Composite Structures*. 119905. [Q1].
4. **Tijani I.A**, T.G Wakjira, M.S Alam, & N Uddin. (2025) Digital image correlation for structural health monitoring of bridge systems: A state-of-the-art review with future research directions. *Archives of Computational Methods in Engineering*. 33, 4325-4341. [Q1].
5. Zhang R, Fares A., **Tijani I.A**, Zayed T. Jin Z, and Yussif A. (2025) Acoustic based leak location detection for water supply pipelines in urban areas via multi-task deep learning. *Engineering Applications of Artificial Intelligence*. 1112718. [Q1].
6. **Tijani I.A**, T.G Wakjira, H. Haroglu & M.S Alam. (2025). A surrogate-based machine learning model for prediction of interfacial fracture energy in externally bonded FRP–concrete bonds with a user-friendly tool. *Engineering Fracture Mechanics*. 111633. [Q1].
7. **Tijani I.A**, T.G Wakjira, H. Haroglu & M.S Alam. (2025) Explainable machine learning and application-oriented tool for predicting effective hoop strain of FRP-confined concrete. *Frontiers of Structural and Civil Engineering*. 19, 1621-1636. [Q1].
8. Wakjira, T.G., **Tijani, I.A**., Alam, M.S., Mashal, M., & Hasan, M.K. (2025) Can we trust AI content detection tools for critical decision-making? *Information*, 16, 904. [Q2].
9. Liu, R., Tariq, S., **Tijani, I.A**, Fan, H, Abdelmageed, S Fares, A., & Zayed, T. (2025). Leak detection in water distribution networks using micro-electromechanical systems-based accelerometers: a machine learning approach. *Water Practice and Technology*. 20(9), 1900-1920. [Q2].
10. **Tijani I.A**, J.A Dauda, M.A Kareem, & A.I Lawal. (2025). Genetic programming-based model for estimating maximum pull load of FRP-to-concrete bond interfaces with GUI implementation. *Structural Concrete*. 26(6), 8899-8912 [Q1].
11. Zhang, R., Yussif, A., **Tijani, I.A**., Fares, A., Tariq, S., & Zayed, T (2024). Acoustic localization approach for urban water distribution networks using machine learning method. *Engineering Applications of Artificial Intelligence*. 137. 109062. [Q1].
12. Liu, R., Tariq, S., **Tijani, I.A**., Fares, A., Bakhtawar, B., Fan, H., Zhang, R., & Zayed, T. (2024). Data-driven approaches for vibroacoustic localization of leaks in water distribution networks. *Environmental Processes.* 11, 14. [Q1].
13. **Tijani, I.A**., Lawal, A.I., Ogunsola, N.O, & Kwon, S. (2023) Prediction of ultimate strength of FRP-confined predamaged concrete using backward multiple regression motivated soft computing methods. *Scientia Iranica*. e23225. [Q3].
14. Kareem, M.A., Akintonde, B.B., Adesoye, J.S., Ishola, K., **Tijani, I.A.**, & others. (2023). Influence of Cashew Leaf Ash as Partial Replacement for Cement on the Properties of Fresh and Hardened Concrete. *Cleaner Waste Systems*. 4, 100063. [Q2].
15. **Tijani, I.A.**, Lawal, A.I., & Kwon, S. (2022). Machine learning techniques for prediction of ultimate strain of FRP-confined concrete. *Structural Engineering and Mechanics* 84(1):101-111 [Q2].
16. **Tijani I.A** & Zayed T. (2022). Gene Expression Programming Based Mathematical Models for Leak Detection of Water Distribution Networks. *Measurement*, 188:110611 [Q1].
17. Dauda, J.A., Rahmon, S., **Tijani, I.A.,** & Mohammed, F. (2022). Design Optimisation of Reinforced Concrete Pile Foundation using Generalized Reduced Gradient Algorithm. *Frontiers in Engineering and Built Environment* 2(3):133-153. [Q2].

18. **Tijani I.A**., Abdelmageed, S., Fares, A., Fan, K.H., Hu, Z.Y., & Zayed, T., (2022). Improving the leak detection efficiency in water distribution networks using noise loggers. *Science of the Total Environment*. 821, 153530. [Q1].
19. Fares, A., **Tijani, I. A** & others (2022) Leak detection in real water distribution networks based on acoustic emission and machine learning. *Environmental Technology*. 44(25), 3850-3866. [Q2].
20. **Tijani I.A**., Jiang, C., Lim. C.W., & Wu, Y.F. (2021). Eccentrically loaded concrete under nonuniform passive confinement. *ASCE Journal Structural Engineering*. 04021247. [Q1].
21. **Tijani I.A**., Wu, Y.F., & Lim, C.W. (2021). "Effects of pre-damage on stress-strain relationship of partially confined concrete." *ACI Structural Journal*. 118(1):61–72. [Q1].
22. **Tijani I.A**., Jiang, C., Lim. C.W., & Wu, Y.F. (2020). Effect of load eccentricity on the mechanical response of FRP-confined pre-damaged concrete columns. *ASCE Journal Composites for Construction*, 04020057. [Q1].
23. **Tijani I.A** & others (2020). "Energy balance method for modeling ultimate strain of fiber-reinforced polymer-repaired concrete." *Structural Concrete*, 21(2):804–820. [Q1].
24. **Tijani I. A** & others (2019). "Aggregate size effects and general static loading response on mechanical behavior of passively confined concrete." *Construction and Building Materials*, 205:61–72. [Q1].